

VxLEARN Networks

Networking & Cybersecurity Track
Simulated Employment Program

Lab Report: Observe Web Requests

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Date: 18 November 2025

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1. Introduction

This lab focuses on understanding how a client interacts with a web server when requesting web services within a network. Although the network used in this activity is small, it demonstrates the same processes that occur every time a user loads a webpage on the internet. The External Client (PC) must first verify its ability to reach the domain **ciscolearn.web.com**, which requires correctly resolving the domain name to an IP address using DNS. Once connectivity is confirmed, the client sends an HTTP request to the server, retrieves the webpage, and displays it in the browser.

Using Packet Tracer's Simulation Mode, this lab allows detailed observation of how DNS queries, HTTP requests, TCP acknowledgments, and webpage data segments travel across the network. By filtering traffic to only show HTTP and TCP events, the packet flow becomes clear and easier to analyze. The goal of this lab is to develop a deeper understanding of the client-server interaction and how the underlying protocols DNS, HTTP, and TCP work together to deliver a webpage successfully.

This lab not only reinforces foundational networking concepts, but also provides hands-on experience analyzing real traffic patterns, making it an essential exercise for building a strong understanding of how websites are accessed and displayed on a client machine.

2. Objectives

- Observe client-server communication when a PC requests a webpage
- Understand how DNS resolves a website name into an IP address
- View the HTTP request and the server's response
- Use simulation mode to analyze traffic flow
- Understand why HTTP creates higher traffic overhead due to TCP

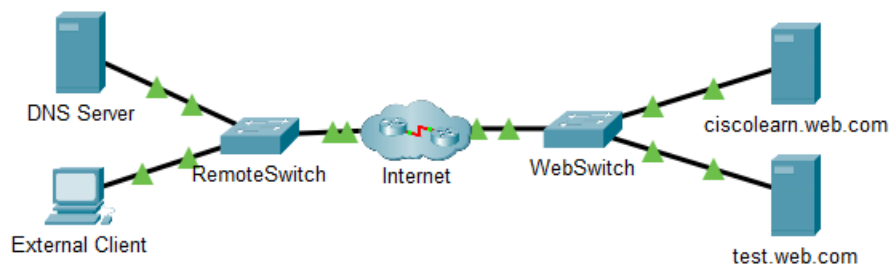
3. Network Overview

This lab uses a simple network containing:

- **External Client (PC)**
- **DNS + Web Server (ciscolearn.web.com)**

The PC must resolve the website name, then contact the server to load the webpage.

This activity focuses on **DNS resolution** and **HTTP webpage retrieval**, showing how these protocols interact.



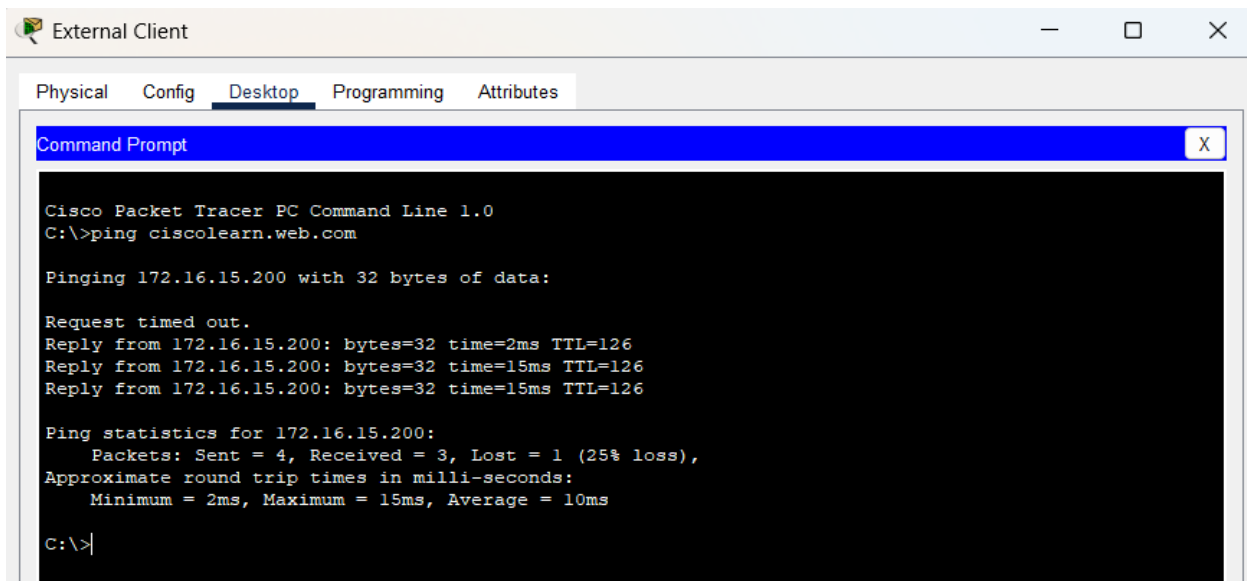
4. Part 1: Verify Connectivity to the Web Server

1. Open the External Client.
2. Go to Desktop > Command Prompt.
3. Enter the command:
`ping ciscolearn.web.com`
4. Observe the DNS-resolved IP address.

The PC cannot ping a name until DNS converts it into an IP address.
That is why the ping output shows something like:

Pinging **192.168.x.x** ... (this IP comes from DNS)

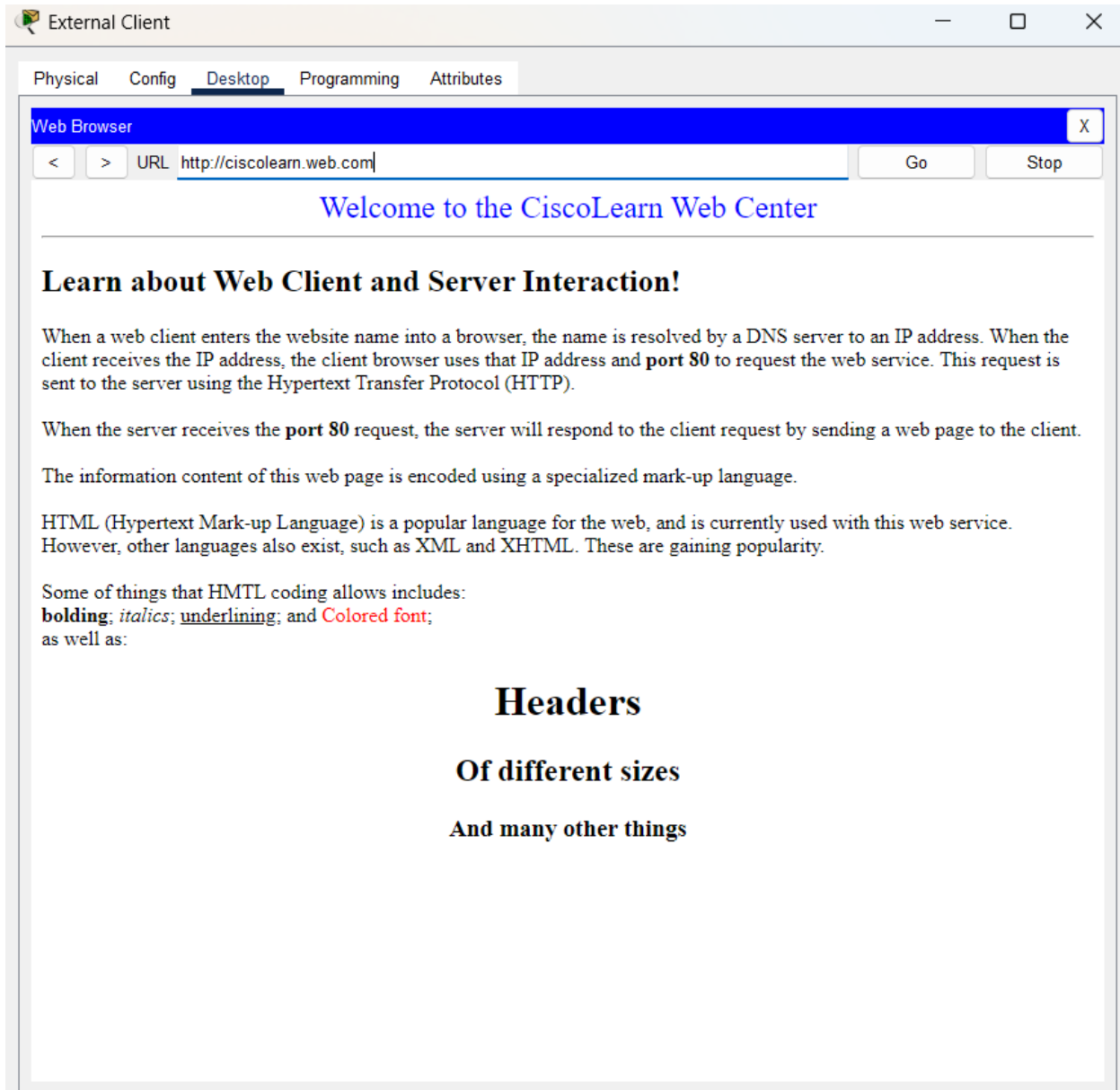
5. Close the Command Prompt (keep External Client window open).



Ping WebServer

5. Part 2: Connect to the Web Server

1. Open Web Browser on the External Client.
2. Enter URL: ciscolearn.web.com
3. Read the displayed webpage.
4. Minimize (do not close) the External Client.



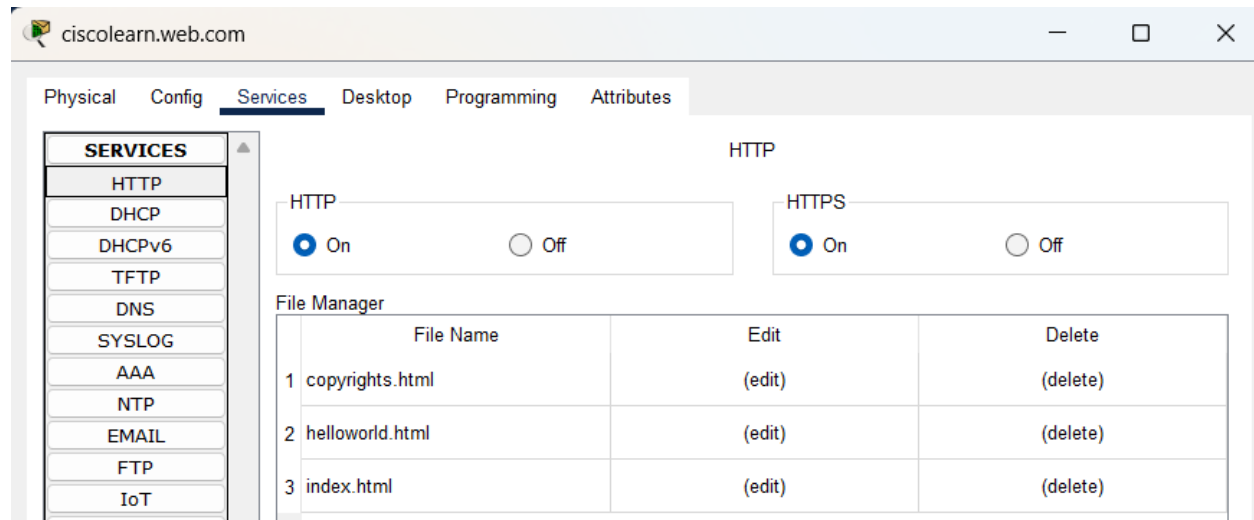
Web Browser loaded

6. Part 3: View HTML Code on the Server

1. Click the ciscolearn.web.com server.

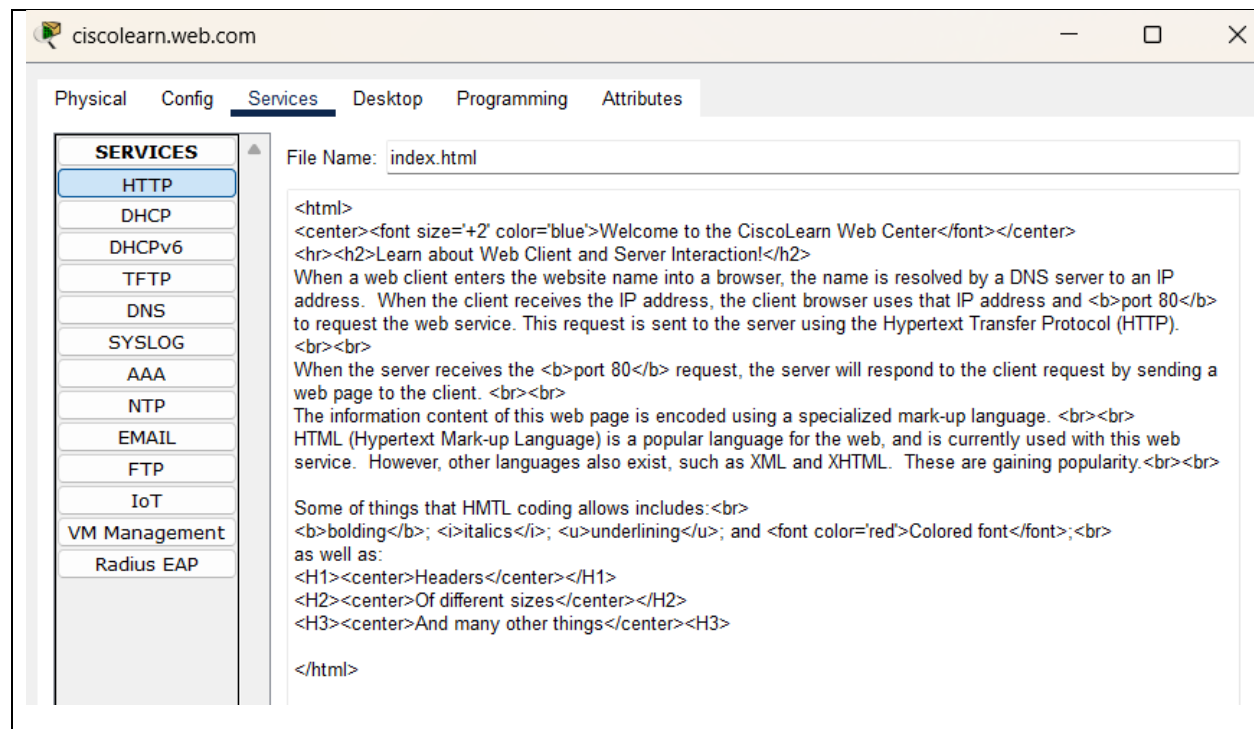


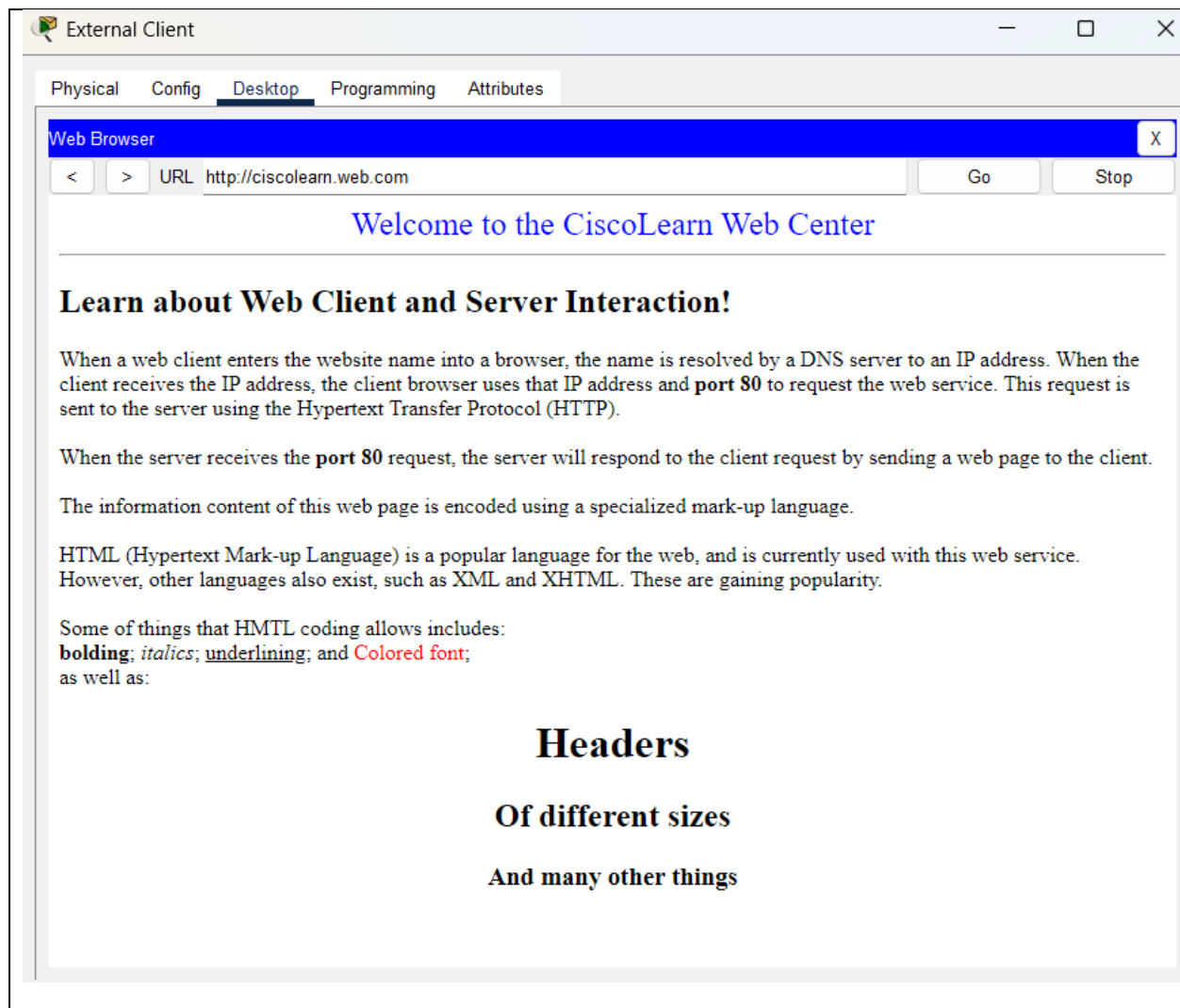
2. Go to Services > HTTP.



3. Click (edit)* beside index.html.

4. Compare the HTML content with the webpage in the browser.

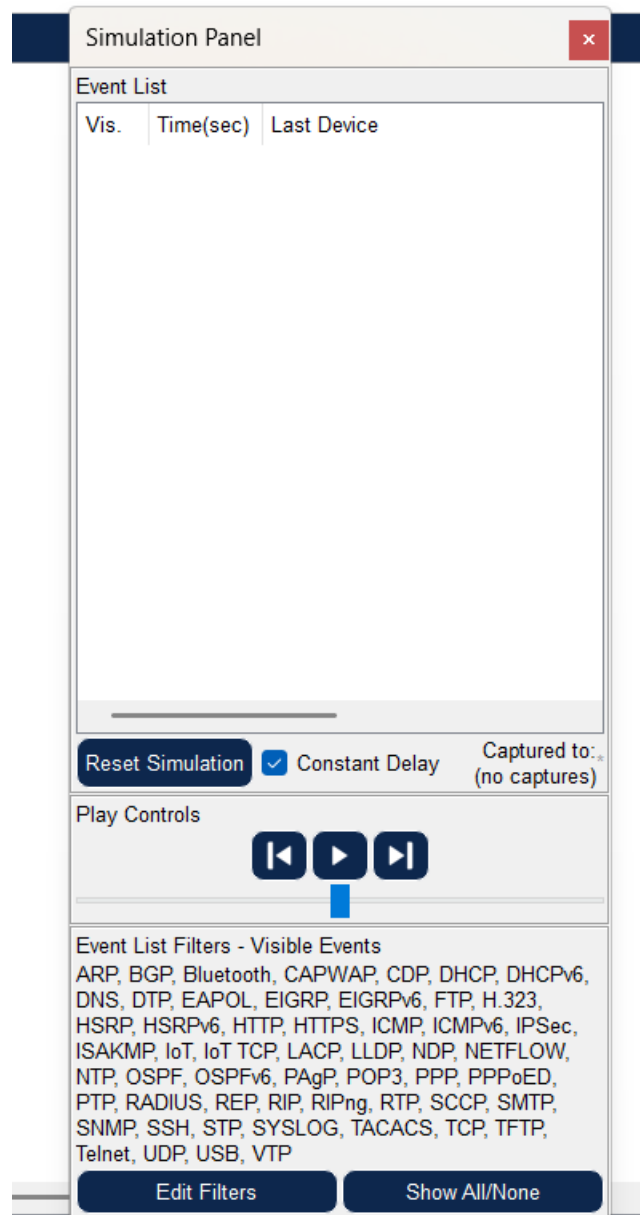




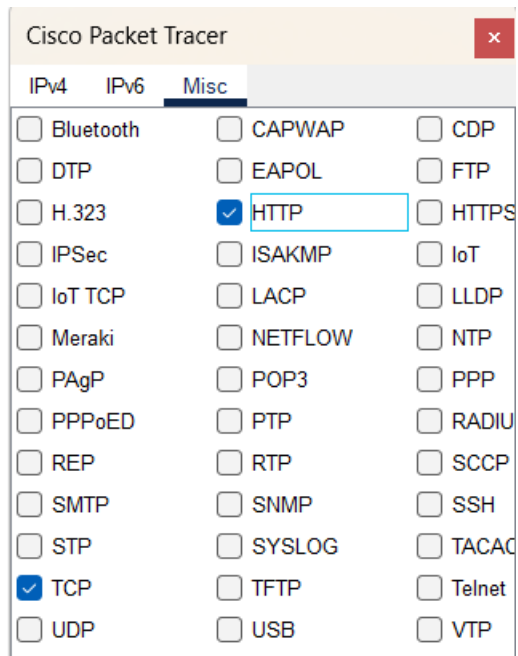
5. Close both windows.

7. Part 4: Observe Traffic Between Client and Server

1. Enter Simulation Mode.
2. Unlock and reposition the Simulation Panel.



3. Open Edit Filters and ensure TCP and HTTP are selected only.



Cisco Packet Tracer

IPv4 IPv6 **Misc**

☐ Bluetooth	☐ CAPWAP	☐ CDP
☐ DTP	☐ EAPOL	☐ FTP
☐ H.323	☒ HTTP	☐ HTTPS
☐ IPSec	☐ ISAKMP	☐ IoT
☐ IoT TCP	☐ LACP	☐ LLDP
☐ Meraki	☐ NETFLOW	☐ NTP
☐ PAgP	☐ POP3	☐ PPP
☐ PPPoED	☐ PTP	☐ RADIUS
☐ REP	☐ RTP	☐ SCCP
☐ SMTP	☐ SNMP	☐ SSH
☐ STP	☐ SYSLOG	☐ TACACS+
☒ TCP	☐ TFTP	☐ Telnet
☐ UDP	☐ USB	☐ VTP

4. Create a Complex PDU:

- Click the open envelope icon.
- Select the External Client as the source.
- Set Application to HTTP.
- Select the server as destination.
- Set Starting Source Port = 1000.
- Set Periodic Interval = 120 seconds.
- Click Create PDU.

Create Complex PDU

Source Settings

Source Device: External Client

Outgoing Port:

FastEthernet0 ☒ Auto Select Port

PDU Settings

Select Application: HTTP

Destination IP Address: 172.16.15.200

Source IP Address:

TTL: 32

TOS: 0

Starting Source Port: 1000

Destination Port: 80

Size: 0

Simulation Settings

☐ One Shot Time: Seconds

☒ Periodic Interval: 120 Seconds

Create PDU

Create Complex PDU

5. Click Play to observe traffic.
6. Use View Previous Events when buffer fills.
7. Scroll the Event List to analyze TCP handshake + HTTP requests.

Simulation Panel

Event List

Vis.	Time(sec)	Last Device
	0.004	Local
	0.005	WebSwitch
	0.005	WebSwitch
	0.006	ciscolearn.web.com
	0.007	WebSwitch
	0.009	Remote
	0.010	RemoteSwitch
	0.010	--
	0.011	External Client
	0.011	--
	0.012	External Client
	0.012	RemoteSwitch
	0.013	RemoteSwitch
	0.014	Local
	0.015	Local
	0.015	WebSwitch
Visible	0.016	WebSwitch

Reset Simulation

☒ Constant Delay

Captured to: 0.016 s

Play Controls

⏮

⏸

⏭

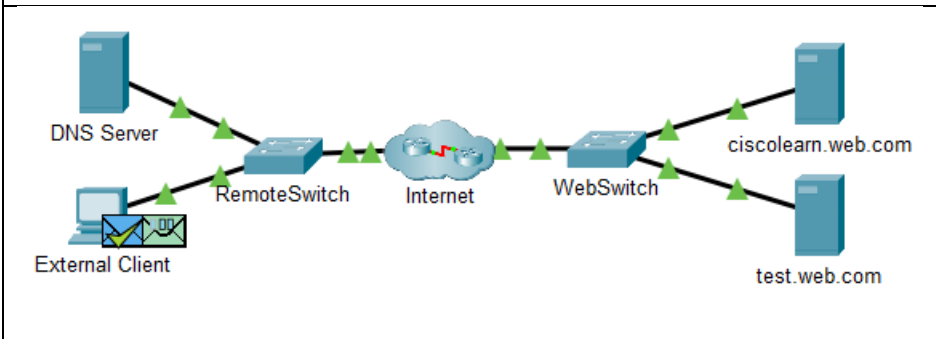
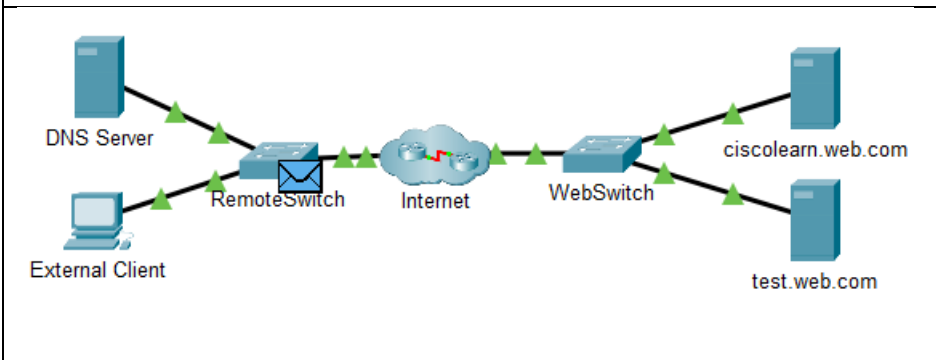
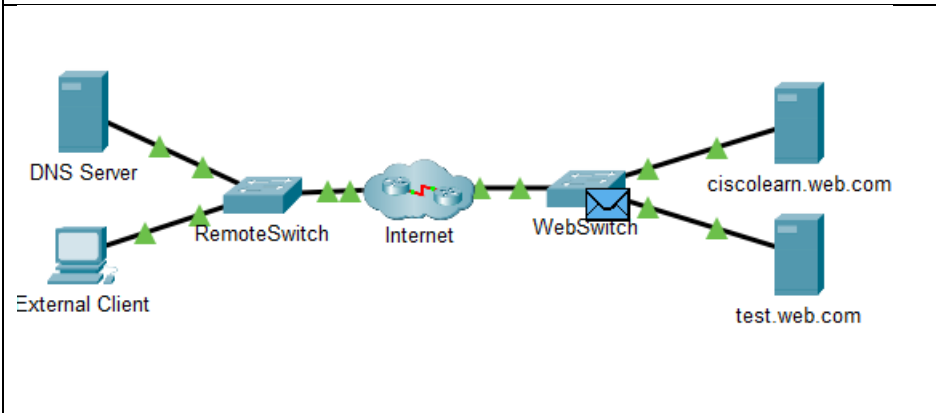
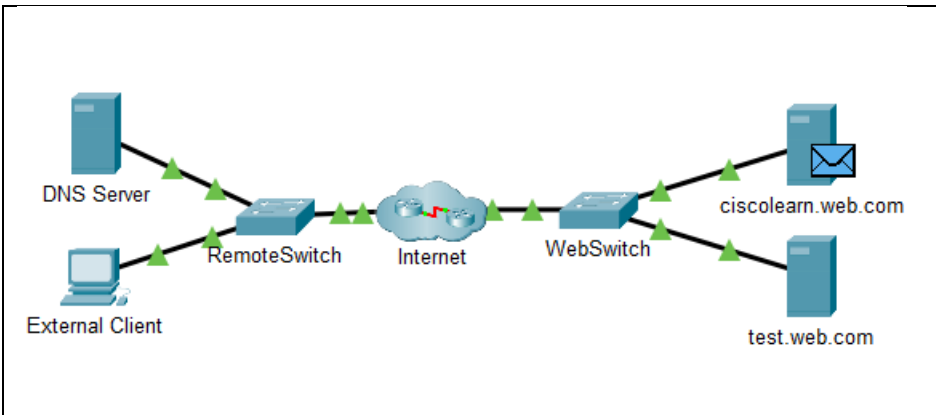
Event List Filters - Visible Events

ARP, BGP, DHCP, DHCPv6, DNS, EIGRP, EIGRPv6, HSRP, HSRPv6, HTTP, ICMP, ICMPv6, NDP, OSPF, OSPFv6, RIP, RIPng, TCP

Edit Filters

Show All/None

Event List Traffic



8. Understanding DNS and HTTP

DNS – How the PC Finds the Server

When I typed **ciscolearn.web.com**:

- The PC did **not** know the IP address
- It sent a DNS request asking:
 “What is the IP of ciscolearn.web.com?”
- The DNS server replied with the correct IP

DNS turns a website name into the numerical IP the computer needs.

Without DNS, the PC would not know where the web server is.

HTTP – How the Webpage Loads

Once the PC had the IP address, it used **HTTP** to request the webpage.

The PC sent:

HTTP GET /index.html

The server responded with the webpage, but **not all at once**.

It sent it in multiple **TCP segments**, such as:

- Segment 1
- Segment 2
- Segment 3

For every segment, the PC sent back a **TCP ACK** (acknowledgment).

Once all segments were received, the browser assembled them and displayed the page.

HTTP is the language used to request webpages.

TCP ensures every part arrives correctly.

9. Question and Answers

1. What IP address did the DNS resolve for ciscolearn.web.com?

172.16.15.200

2. Why does the HTTP request generate many packets?

Because HTTP uses TCP, which requires a connection handshake and acknowledgment for each packet.

Webpages are larger than a single packet, so they are split into multiple segments.

3. What did the HTML code show?

It showed the raw markup code that the server uses to generate the webpage.

10. Conclusion

This lab demonstrated:

- How DNS resolves a website name to an IP address
- How HTTP retrieves a webpage
- How TCP ensures reliable communication
- How Packet Tracer simulation mode visualizes this traffic

Understanding these steps shows how client–server interactions work and why web browsing involves multiple protocols working together.